All Skills Combined

Core AI/ML

- Transformer models (BERT, GPT, LLaMA) and attention mechanisms
- Fine-tuning (LoRA, PEFT) and prompt engineering
- RAG pipeline design
- AI agents with memory, planning, and tool use
- Multi-agent systems
- Model compression, optimization, and evaluation (BLEU, ROUGE)
- Embeddings, vector databases, semantic search

LLM Frameworks & Tooling

- LangChain, LlamaIndex
- Hugging Face Transformers
- OpenAI / Anthropic APIs
- LLMOps: monitoring, tracing, prompt tracking, hallucination mitigation

NLP

- Intent detection, sentiment analysis, named entity recognition
- Dialogue management and state tracking
- Text classification

ML Frameworks

- PyTorch (primary), TensorFlow
- Pandas, NumPy

Backend & APIs

- Python (advanced, async)
- FastAPI, Flask, Django
- API design and security

Databases

- SQL and advanced SQL
- PostgreSQL, MongoDB
- Snowflake, SingleStore, BigQuery
- Vector databases (Pinecone, Vertex Vector Search, ChromaDB)
- ORM (SQLAlchemy)
- Dimensional data modeling

Data Engineering

- Big data processing at scale

- Feature engineering and data pipelines
- Automated ML pipelines

Cloud & DevOps

- AWS (primary), GCP, Azure
- Docker, Kubernetes
- CI/CD pipelines
- Infrastructure as Code (Terraform)
- Serverless (Lambda, Cloud Functions)
- LLMOps pipelines

Soft Skills

- Breaking down business problems into AI solutions
- Cross-functional collaboration
- End-to-end ownership (dev to production)
- Governance, security, compliance

Learning Roadmap (Start Here, Go in Order)

Think of this as building a house. You need the foundation before the walls, walls before the roof.

Stage 1: The Foundation (2-3 months)

Start here. Everything else depends on this.

- **Python**: Get genuinely good at it. Learn functions, classes, file handling, async programming, and libraries like Pandas and NumPy. This is non-negotiable.
- **SQL**: Learn how to query, join, filter, and aggregate data. Then move to window functions and CTEs.
- **Basic statistics and linear algebra**: Means, distributions, matrix multiplication. You don't need to go deep, just enough to understand what models are doing.

Resources: Python.org docs, W3Schools SQL, Khan Academy for math.

Stage 2: Machine Learning Basics (2-3 months)

Before LLMs, understand what ML actually is.

- How models learn (gradient descent, loss functions)
- Classification, regression, clustering
- Train/test splits, overfitting, evaluation metrics
- PyTorch basics: tensors, building a simple neural network

Resources: fast.ai (free, practical), Andrej Karpathy's YouTube channel.

Stage 3: Understand How LLMs Work (1-2 months)

This is where it gets exciting.

- What transformers are and how attention works (read "Attention Is All You Need" after watching explanations)
- How GPT-style models are trained
- Tokenization, embeddings, context windows
- Karpathy's "Let's build GPT from scratch" video on YouTube is gold here

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Stage 4: Working With LLMs (2-3 months)

Now you start using them, not just understanding them.

- OpenAI and Anthropic APIs: send prompts, get responses, handle conversations
- Prompt engineering: zero-shot, few-shot, chain-of-thought
- Hugging Face: load and run open-source models locally
- Fine-tuning a small model using LoRA/PEFT
- Evaluation: BLEU, ROUGE, and manual review

Stage 5: Building AI Applications (2-3 months)

This is where you go from experiments to real products.

- **LangChain and LlamaIndex**: build chains, manage memory, handle multi-turn conversations
- **RAG pipelines**: connect LLMs to your own data using vector databases (start with ChromaDB or FAISS, then Pinecone)
- **FastAPI**: build a simple API that serves your LLM app
- Build a small project end to end, something like a document Q&A bot

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Stage 6: AI Agents and Multi-Agent Systems (1-2 months)

- What agents are and how they use tools
- Building agents with LangChain or LlamaIndex
- Planning, memory, and execution loops
- Multi-agent coordination (AutoGen, CrewAI)

Stage 7: Cloud, DevOps, and Production (2-3 months)

This separates engineers who prototype from those who ship.

- **Docker**: containerize your app
- **AWS or GCP**: deploy your container, use S3/GCS for storage, Lambda for serverless
- **CI/CD basics**: GitHub Actions to automate testing and deployment
- **LLMOps**: monitor your model in production, track prompts, catch hallucinations (tools: LangSmith, Weights & Biases)

Stage 8: Data Engineering and Scale (ongoing)

This is more relevant for senior roles like Apple's.

- Big data tools (Spark, dbt)
- Snowflake or BigQuery
- Building data pipelines
- Dimensional data modeling